

ECOM30001/90001 Basic Econometrics

Tutorial 10: Simultaneous Equations Models (SEM) & 2SLS

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Tutorial 10 Overview

Today's Core Goals

- **Simultaneous Equations Bias:** Why OLS fails when variables are jointly determined in equilibrium.
- **Identification in SEM:** Structural form vs. reduced form, order condition, and exclusion restrictions.
- **Practical Identification:** Testing instrument relevance via reduced-form F -tests (Wald tests).
- **Question 1 (Fulton Fish):** Demand vs. Supply, weather instruments, and price elasticity test.
- **Question 2 (Labour Supply):** Joint determination of hours and wages for partnered women.
- **Exam Practice MCQ:** System identification under zero restrictions.

Core Concepts: Simultaneity & Identification

1. The Simultaneity Problem (Why OLS is Inconsistent)

When P and Q are jointly determined by equilibrium:

- A shock ε_{dt} affects $Q_t \implies$ alters market clearing price P_t .
- Therefore, $\text{Cov}(\ln P_t, \varepsilon_{dt}) \neq 0$ and $\text{Cov}(\ln P_t, \varepsilon_{st}) \neq 0$.
- **Result:** OLS estimates are biased and inconsistent for structural parameters.

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- **Result:** OLS estimates are biased and inconsistent for structural parameters.

2. Conditions for Identification

- **Theoretical Identification (Order Condition):** Requires enough exogenous variables **excluded** from the equation of interest and **included** in the rest of the system ($K_{\text{excluded}} \geq M_{\text{included}} - 1$). >= n-1
- **Practical Identification (Rank Condition):** The excluded instruments must have **statistically significant** explanatory power in the first stage ($F > 10$).

Question 1: Fulton Fish Market Setup

Econometric Model: Demand and Supply for Whiting

Demand: $\ln Q_t = \alpha_0 + \alpha_1 \ln P_t + \alpha_2 \text{MON}_t + \alpha_3 \text{TUES}_t + \alpha_4 \text{WED}_t + \alpha_5 \text{THU}_t + \varepsilon_{dt}$

Supply: $\ln Q_t = \beta_0 + \beta_1 \ln P_t + \beta_2 \text{STORMY}_t + \varepsilon_{st}$

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Variable Classification & Logic:

- **Endogenous variables** ($M = 2$): $\ln Q_t$ (quantity), $\ln P_t$ (price).
- **Exogenous variables** ($K = 5$): Weekday dummies ($\text{MON}_t, \text{TUES}_t, \text{WED}_t, \text{THU}_t$) and STORMY_t (weather indicator).
- **Economic Intuition:** Stormy weather reduces fishing catch (shifts supply leftward, $\beta_2 < 0$), while day-of-week captures buyer shopping patterns (shift demand).

Question 1(a) & (b): Theoretical Identification

Part (a): Is the Demand Equation Identified?

YES!

- **Excluded Exogenous Variable:** $STORMY_t$ is excluded from demand.
- $STORMY_t$ shifts the supply curve, tracing out the demand curve.
- *Order Condition:* Excluded exogenous variables (1) $\geq$ Included endogenous regressors (1) $\implies$ **Exactly Identified.**

n-1

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Part (b): Is the Supply Equation Identified?

YES!

- **Excluded Exogenous Variables:** $\text{MON}_t, \text{TUES}_t, \text{WED}_t, \text{THU}_t$ are excluded from supply.
- Weekday dummies shift the demand curve, tracing out the supply curve.
- *Order Condition:* Excluded exogenous variables (4) $\geq$ Included endogenous regressors (1) $\implies$ **Overidentified.**

Question 1(c): Reduced Form Estimates

Reduced Form Specifications (Estimated by OLS)

$$\ln Q_t = \pi_{Q0} + \pi_{Q1}\text{MON}_t + \pi_{Q2}\text{TUES}_t + \pi_{Q3}\text{WED}_t + \pi_{Q4}\text{THU}_t + \pi_{Q5}\text{STORMY}_t + v_{Qt}$$

$$\ln P_t = \pi_{P0} + \pi_{P1}\text{MON}_t + \pi_{P2}\text{TUES}_t + \pi_{P3}\text{WED}_t + \pi_{P4}\text{THU}_t + \pi_{P5}\text{STORMY}_t + v_{Pt}$$

Regressor	ln(Q) Reduced Form		ln(P) Reduced Form	
	Estimate (Robust SE)	p-value	Estimate (Robust SE)	p-value
(Intercept)	8.8101*** (0.1174)	< 0.001	-0.2717** (0.0952)	0.0052
mon	0.1010 (0.1978)	0.6106	-0.1129 (0.1154)	0.3302
tue	-0.4847* (0.1939)	0.0140	-0.0411 (0.1164)	0.7245
wed	-0.5531** (0.1986)	0.0063	-0.0118 (0.1135)	0.9173
thu	0.0537 (0.1720)	0.7555	0.0496 (0.1156)	0.6683
stormy	-0.3878** (0.1426)	0.0076	0.3464*** (0.0723)	< 0.001

Question 1(c): Testing Practical Identification

(c)(i) Supply Equation: Is it Practically Identified?

NO! Requires weekday variables to be jointly significant in $\ln P_t$ reduced form:

$$H_0 : \pi_{P1} = \pi_{P2} = \pi_{P3} = \pi_{P4} = 0 \implies \mathbf{F(4, 105) = 0.7426, p = 0.5651}$$

► Fail to reject H_0 ($p = 0.5651 > 0.05$). Weekdays do not shift price $\implies$
Supply equation is NOT practically identified!

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Supply equation is NOT practically identified!

(c)(ii) Demand Equation: Is it Practically Identified?

YES! Requires STORMY_t to be statistically significant in $\ln P_t$ reduced form:

$$H_0 : \pi_{P5} = 0 \implies \mathbf{F(1, 105) = 22.929, p = 0.0000055 \quad (t = 4.79)}$$

► Reject H_0 ($p < 0.001, F = 22.93 \gg 10$). Stormy weather strongly shifts supply and moves price $\implies$ **Demand is practically identified!**

Question 1(d): 2SLS Demand Estimation (Fulton Fish)

2SLS Estimation via `ivreg()` (Robust SEs, $N = 111$)

Regressor	Estimate	Robust SE	t value	$\text{Pr}(> t)$
(Intercept)	8.5059***	0.1521	55.93	< 0.001
$\ln(\text{price})$ ($\hat{\alpha}_1$)	-1.1194*	0.4432	-2.53	0.0130
mon	-0.0254	0.2214	-0.11	0.9089
tue	-0.5308**	0.2021	-2.63	0.0099
wed	-0.5664**	0.2069	-2.74	0.0073
thu	0.1093	0.1784	0.61	0.5416

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Key Takeaway from 2SLS Demand Estimates

- The 2SLS coefficient on ln(price) is statistically significant ($\hat{\alpha}_1 = -1.1194$, $p = 0.0130$).
- Significant day-of-week effects on Tuesday and Wednesday indicate lower baseline fish demand mid-week relative to Friday.

Question 1(d): Price Elasticity & Hypothesis Testing

1. Economic Interpretation of $\hat{\alpha}_1$

In this log-log demand specification:

$$\hat{\alpha}_1 = \frac{\partial \ln Q}{\partial \ln P} = \text{own-price elasticity of demand} = -\mathbf{1.1194}$$

► All else equal, a **1% increase in price** reduces quantity demanded by **1.12%**.

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2. Formal Hypothesis Test: Is Demand Price Elastic?

Demand is price elastic if $\alpha_1 < -1$. We formulate the one-tailed test:

$$H_0 : \alpha_1 \geq -1 \quad \text{vs.} \quad H_1 : \alpha_1 < -1$$

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Test Statistic:

$$t = \frac{\hat{\alpha}_1 - (-1)}{\text{SE}(\hat{\alpha}_1)} = \frac{-1.119417 - (-1)}{0.443193} = -0.2694$$

p-value: $\Pr(t_{105} \leq -0.2694) = 0.394 > 0.05$.

Conclusion: Fail to reject H_0 . Despite a point estimate of -1.12 , there is no statistical evidence that demand is significantly more elastic than unitary elasticity.

Question 2: Female Labour Supply & Inverse Demand

Structural Econometric Model for Partnered Women ($N = 2,867$)

Inverse Demand: $\ln(\text{wage}_i) = \alpha_0 + \alpha_1 \text{hours}_i + \alpha_2 \text{educ}_i + \alpha_3 \text{exper}_i + \alpha_4 \text{exper}_i^2 + \alpha_5 \text{union}_i + \varepsilon_i$
(1)

Labour Supply: $\text{hours}_i = \beta_0 + \beta_1 \ln(\text{wage}_i) + \beta_2 \text{educ}_i + \beta_3 \text{childlt6}_i + \beta_4 \text{childge6}_i + \beta_5 \text{nlr}_i + \eta_i$
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(2)

System Structure:

- **Endogenous variables** ($M = 2$): $\ln(\text{wage}_i)$ and hours_i (jointly determined).
- **Exogenous in Demand only** ($K_1 = 3$): $\text{exper}_i, \text{exper}_i^2, \text{union}_i$ (productivity & union shifters).
- **Exogenous in Supply only** ($K_2 = 3$): $\text{childlt6}_i, \text{childge6}_i, \text{nlincl}_i$ (household demographics & non-labour income).
- **Exogenous in Both:** educ_i (education affects both demand productivity and supply preferences).

Question 2(a): Practical Identification of Labour Demand (1)

Conditions for Identification of Labour Demand Equation (1)

Labour demand (1) excludes: childlt6_i , childge6_i , nlinc_i .

► Requires these 3 variables to be jointly significant in the **reduced form for hours_i** :

$$H_0 : \pi_{h5} = \pi_{h6} = \pi_{h7} = 0$$

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Relevant Test (Figure 1)	Res. Df	Df	F-statistic	Pr(> F)
$H_0 : \text{kidlt6} = \text{kidge6} = \text{nlinc} = 0$	2859	3	30.247	< 0.001

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Conclusion

Reject H_0 ($F = 30.25 \gg 10, p < 0.001$). Demographics strongly shift labour supply $\Rightarrow$ **Labour demand equation (1) is practically identified!**

Why not Figure 2? Figure 2 erroneously tests $\text{educ} = 0$, but education is included in both equations!

Question 2(b): Practical Identification of Labour Supply (2)

Conditions for Identification of Labour Supply Equation (2)

Labour supply (2) excludes: $\text{exper}_i, \text{exper}_i^2, \text{union}_i$.

► Requires these 3 variables to be jointly significant in the **reduced form** for $\ln(\text{wage}_i)$:

$$H_0 : \pi_{w2} = \pi_{w3} = \pi_{w4} = 0$$

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Relevant Test (Figure 4)	Res. Df	Df	F-statistic	Pr(> F)
$H_0 : \text{exper} = \text{exper}^2 = \text{union} = 0$	2859	3	50.925	< 0.001

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Relevant Test (Figure 4)	Res. Df	Df	F-statistic	Pr(> F)
$H_0 : \text{exper} = \text{exper}^2 = \text{union} = 0$	2859	3	50.925	< 0.001

Conclusion

Reject H_0 ($F = 50.93 \gg 10, p < 0.001$). Experience and union status strongly shift labour demand $\Rightarrow$ **Labour supply equation (2) is practically identified!**
Why not Figure 3? Figure 3 erroneously tests $\text{educ} = 0$, but education is included in both equations!

Question 2(c): 2SLS Labour Supply Estimates

2SLS Regression Results for Model (2) (Figure 5, $N = 2,867$)

Regressor	Estimate	Robust SE	t value	$\Pr(> t)$
(Intercept)	23.7588***	2.5521	9.31	< 0.001
$\ln(\text{wage}) (\beta_1)$	9.6186***	2.0806	4.62	< 0.001
$\text{educ} (\beta_2)$	-0.4213	0.2330	-1.81	0.0707
$\text{childlt6} (\beta_3)$	-3.0832***	0.6169	-5.00	< 0.001
$\text{childge6} (\beta_4)$	-2.1793***	0.5028	-4.33	< 0.001
$\text{nlinc} (\beta_5)$	-0.0690***	0.0115	-6.00	< 0.001
$F(5, 2861) = 20.27$	$p < 0.001$	Residual SE: 11.4059		

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$F(5, 2861) = 20.27$ $p < 0.001$ Residual SE: 11.4059				

Main Estimation Highlights

- All family composition and non-labour income variables are statistically significant at the 1% level ($p < 0.001$).
- Wage coefficient is positive and significant ($\hat{\beta}_1 = 9.62, p < 0.001$).

Question 2(c): Economic Interpretation of Parameters

1. Young Child Coefficient ($\hat{\beta}_3 = -3.0832$)

- **Interpretation:** Partnered women with a child aged under 6 years at home work **on average approx 3.08 fewer hours per week compared to** women without a young child, **holding** wage rate, education, older children, and non-labour income constant.
- **Economic Logic:** High childcare burden significantly constrains available market labour time.

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2. Additional Economic Insights

- **Older Children** ($\hat{\beta}_4 = -2.1793$): Having a child aged ≥ 6 reduces weekly hours by ≈ 2.18 hours (less restrictive than infant/toddler care).
- **Non-Labour Income** ($\hat{\beta}_5 = -0.0690$): For each \$1,000 increase in family non-labour income, weekly hours decrease by 0.069 hours $\implies$ **Income effect: leisure is a normal good.**
- **Wage Elasticity** ($\hat{\beta}_1 = 9.6186$): Confirms an **upward-sloping labour supply curve.**

Exam Practice Question (MCQ)

Exam Problem: Demand & Inverse Supply Identification

Consider the log-linear SEM:

$$q_t = \alpha_0 + \alpha_1 p_t + \alpha_2 x_t + \alpha_3 z_t + \varepsilon_{1t} \quad (\text{demand})$$

$$p_t = \beta_0 + \beta_1 q_t + \beta_2 x_t + \beta_3 z_t + \varepsilon_{2t} \quad (\text{inverse supply})$$

where x_t and z_t are exogenous variables that potentially affect demand and/or supply.

Question: If parameter restrictions are $\alpha_2 = \beta_2 = \beta_3 = 0$, then:

- ① both demand and (inverse) supply equations are identified
- ② the demand equation is identified but the (inverse) supply equation is not
- ③ the (inverse) supply equation is identified but the demand equation is not
- ④ none of the above

Exam Practice Question: Solution & Explanation

Correct Answer: Option 3 (only inverse supply is identified)

Under the restrictions $\alpha_2 = \beta_2 = \beta_3 = 0$, the system reduces to:

$$\text{Demand: } q_t = \alpha_0 + \alpha_1 p_t + \alpha_3 z_t + \varepsilon_{1t}$$

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Step-by-Step Identification Analysis:

- **Demand Equation:** Excludes x_t , but x_t does not appear in the supply equation either. There is **no exogenous variable** shifting supply to trace out demand $\implies$ **Demand is NOT identified.**
- **Inverse Supply Equation:** Excludes z_t , and z_t appears in the demand equation (shifts demand). Thus, z_t serves as an instrument for $q_t \implies$ **Inverse supply is identified!**

Summary & Looking Ahead

Key Takeaways from Tutorial 10

- **Simultaneity $\implies$ OLS Inconsistency:** When variables are jointly determined in equilibrium, OLS fails.
- **Identification Rule:** To identify an equation for Y_1 , you need an exogenous variable that is **excluded** from Y_1 's equation and **shifts** the other equation.
- **Practical Identification:** Always check the reduced-form F -statistic / Wald test for joint significance ($F > 10$).
- **Estimation:** Use `ivreg(y ~ endogenous + exogenous_included | exogenous_all, data = ...)`.

Thank you for your active participation!